2023 SSMO Team Round Solutions

SMO Team

TR 1: Let (a, b, c, d) be a permutation of $(2, 0, 2, 3)$. Find the largest possible value of $a^b + b^c + c^d + d^a$.

Answer: $\boxed{18}$

Solution: Since the expression is unchanged when (a, b, c, d) is cyclically shifted, WLOG assume $a = 0$. Then (a, b, c, d) is one of $(0, 2, 2, 3), (0, 2, 3, 2), (0, 3, 2, 2)$. The corresponding values of $a^b + b^c + c^d + d^a$ are $13, 18, 14$, so the maximum is $\boxed{18}$.

TR 2: A plane and a car start moving at the same time. The car moves northbound at 60 miles per hour. The plane moves northeast and increases in altitude at an angle of 30°. Let s be the speed in feet per second that the plane must fly at to move north at the same speed as the car. Find $3s^2$.

Answer: $\boxed{61952}$

Solution: Note that 60 miles per hour is $\frac{60 \cdot 5280}{3600} = 88$ feet per second. Since the plane climbs at an angle of 30°, its horizontal speed is $\frac{\sqrt{3}}{2}s$. Since it moves northeast, the northward component of this is

$$\frac{\sqrt{3}}{2}s \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{6}}{4}s.$$

So,

$$\frac{\sqrt{6}}{4}s = 88 \implies s^2 = \frac{16 \cdot 88^2}{6},$$

giving an answer of $3s^2 = 8 \cdot 88^2 = \boxed{61952}$.

TR 3: Let ABC be a triangle such that $AB = 4\sqrt{2}$, $BC = 5\sqrt{2}$, and $AC = \sqrt{82}$. Let ω be the circumcircle of $\triangle ABC$. Let D be on the circle such that $\overline{BD} \perp \overline{AC}$. Let E be the point diametrically opposite B. Let F be the point diametrically opposite D. The area of the quadrilateral $ADEF$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{1221}$

Solution:

Note that $\triangle ABC$ is right by the Pythagorean Theorem. Let O be the midpoint of AC (the circumcenter) and let T be the foot from B to AC, which is also the midpoint of BD. Since $OA = OD$, the distance from A to line DF equals the distance from D to line AC, which

is BT. So, $[ADF] = \frac{1}{2} \cdot DF \cdot BT = \frac{1}{2} \cdot AC \cdot BT = [ABC]$. Since BE and DF are both diameters, $BDEF$ is a rectangle, so $[BDEF] = 4[BOD] = 8[BTO]$. We have

$$[ABC] = \tfrac{1}{2} \cdot 5\sqrt{2} \cdot 4\sqrt{2} = 20.$$

Thus,

$$[ADEF] = [ADF] + [DEF] = [ABC] + 4[BTO].$$

Since $\triangle ABC \sim \triangle ATB$, we get

$$AT = AB \cdot \frac{AB}{AC} = \frac{32}{\sqrt{82}},$$

so

$$TO = \frac{\sqrt{82}}{2} - \frac{32}{\sqrt{82}} = \frac{9}{\sqrt{82}}$$

and

$$BT = \frac{40}{\sqrt{82}}.$$

Hence,

$$[BTO] = \tfrac{1}{2} \cdot \frac{9}{\sqrt{82}} \cdot \frac{40}{\sqrt{82}} = \frac{90}{41}.$$

Therefore,

$$[ADEF] = 20 + 4 \cdot \frac{90}{41} = \frac{1180}{41} \implies 1180 + 41 = \boxed{1221}.$$

TR 4: Find the sum of all primes p such that $p \mid (2023^{p^2} + (p-1)! + 2^{p^4})$.

Answer: $\boxed{36}$

Solution: From Fermat's Little Theorem, we have

$$a^p \equiv a \pmod{p},$$

and from Wilson's Theorem, we have $(p-1)! \equiv -1 \pmod{p}$. Thus,

$$2023^{p^2} + (p-1)! + 2^{p^4} \equiv 2023 - 1 + 2 \equiv 2024 \pmod{p},$$

meaning that $p \mid 2024$. This holds for $p = 2, 11, 23$, giving an answer of

$$2 + 11 + 23 = \boxed{36}.$$

TR 5: Joshy is playing a game with a dartboard that has two sections. If Joshy hits the first section, he gets 20 points, and if he hits the second section, he gets 23 points. Assume Joshy always hits one of the two sections. Let a be the largest score Joshy cannot achieve. Let b be the number of positive integer scores Joshy cannot achieve. Let c be the number of ordered pairs (x, y) of nonnegative integers with $20x + 23y = 2023$. Find $(a - b)c$.

Answer: $\boxed{1040}$

Solution: From the Chicken McNugget Theorem, we have $a = 20 \cdot 23 - 20 - 23 = 417$ and $b = \frac{(20-1)(23-1)}{2} = 209$, so $a - b = 208$. Now, the solution set of $20x + 23y = 2023$ is given by

$$x = 23n + 8 \quad \text{and} \quad y = 81 - 20n,$$

so n can take values from 0 to 4, giving $c = 5$. Thus,

$$(a - b)c = 208 \cdot 5 = \boxed{1040}.$$

TR 6: Suppose that a, b, and c are positive reals satisfying

$$(a^3+4)(b^3+6)(c^3+8) = 8(a+b+c)^3.$$

Find the sum of all possible values of $\frac{bc}{a^2}$. (If there are none, give an answer of 0.)

Answer: $\boxed{0}$

Solution: From Hölder's Inequality, we have

$$(a^3+2+2)(2+b^3+4)(4+4+c^3) \geq 8(a+b+c)^3.$$

Equality holds when $a^3 : 2 : 2 = 2 : b^3 : 4 = 4 : 4 : c^3$. From the first and third ratios, we obtain $a^3 = 2$. However, from the first and second ratios, we obtain $a^3 = 1$, a contradiction. Thus, there are no solutions, and the answer is $\boxed{0}$.

TR 7: Let $S = \{1, 2, 3, 4, \ldots, 23\}$ and let sets $A, B, C \subseteq S$ be chosen independently and uniformly at random. The probability that $|A| + |B| = |C|$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Let 2^a be the largest power of 2 such that $2^a \mid n$. Find a.

Answer: $\boxed{64}$

Solution: The number of ordered triples (A, B, C) satisfying the condition is

$$\sum_{\substack{i,j\geq 0 \\ i+j\leq 23}} \binom{23}{i}\binom{23}{j}\binom{23}{i+j} = \sum_{\substack{i,j\geq 0 \\ i+j\leq 23}} \binom{23}{i}\binom{23}{j}\binom{23}{23-i-j}.$$

Letting $k = 23 - i - j$, this is equivalent to

$$\sum_{i+j+k=23} \binom{23}{i}\binom{23}{j}\binom{23}{k}.$$

By the binomial theorem, we have

$$(x+1)^{23} = \sum_{i=0}^{23} \binom{23}{i} x^i,$$

so the expression is the coefficient of x^{23} in $(x+1)^{69} = ((x+1)^{23})^3$, which is $\binom{69}{23}$. The total number of possible ordered triples (A, B, C) is $(2^{23})^3 = 2^{69}$, so the probability is

$$\frac{\binom{69}{23}}{2^{69}}.$$

Since $\binom{69}{23} = \frac{69!}{23!\,46!}$, applying Legendre's formula shows that the largest power of 2 dividing it is 2^5. Therefore,

$$69 - 5 = \boxed{64}.$$

TR 8: Three rabbits run away from the origin at the same speed and constant velocity such that the angle between any two rabbits' directions is 120°. After 12 seconds, a hunter with a speed $\sqrt{7}$ times that of the rabbits runs from the origin. The minimal time in seconds, measured from when the hunter starts running, needed for her to catch all three rabbits

can be expressed as $a + b\sqrt{c}$, where a, b, and c are positive integers and c is not divisible by the square of any prime. Find $a + b + c$.

Answer: $\boxed{59}$

Solution: Let the rabbits run at speed 1, and let t denote the time since the rabbits started. Running straight to each interception point catches each rabbit as early as possible. When the hunter catches a rabbit at time t_0, the other rabbits are also at distance t_0 from the origin. Suppose she catches the next rabbit τ seconds later. Since the angle between the two rabbits' paths is 120°, from the Law of Cosines, we have

$$\begin{aligned} 7\tau^2 &= t_0^2 + (t_0+\tau)^2 + t_0(t_0+\tau) \implies \\ 0 &= 2\tau^2 - t_0\tau - t_0^2 = (2\tau + t_0)(\tau - t_0), \end{aligned}$$

so $\tau = t_0$. This means each chase after the first takes as long as the time elapsed so far. The hunter catches the first rabbit when

$$\sqrt{7}(t_1 - 12) = t_1 \implies t_1 = \frac{12\sqrt{7}}{\sqrt{7}-1} = 14 + 2\sqrt{7}.$$

So, she catches the second rabbit at time $2t_1$ and the third rabbit at time $4t_1 = 56 + 8\sqrt{7}$, which is $44 + 8\sqrt{7}$ seconds after she starts. Thus, the answer is $44 + 8 + 7 = \boxed{59}$.

TR 9: Let B, K, and R be the total number of possible moves for a bishop, knight, or rook, respectively, summed over all positions of a 9 by 9 grid. Find $B + K + R$. (A bishop moves along diagonals, a rook moves along rows and columns, and a knight moves in the form of an L: two squares in one direction and one square in a perpendicular direction.)

Answer: $\boxed{2560}$

Solution: Define an (i, j)-move as a move that goes i squares to the right and j squares up. Then there are exactly $(9 - |i|)(9 - |j|)$ ways to perform such a move. A rook can perform moves of the form $(0, i)$ and $(i, 0)$ for integers $i \neq 0$ with $1 \leq |i| \leq 8$, so

$$R = 4 \cdot \sum_{i=1}^{8} 9(9-i) = 4 \cdot 9 \cdot \frac{8 \cdot 9}{2} = 1296.$$

A bishop can perform moves of the form (i, i) and $(-i, i)$ for $1 \leq |i| \leq 8$, so

$$B = 4 \cdot \sum_{i=1}^{8} (9-i)^2 = 4 \cdot \frac{8 \cdot 9 \cdot 17}{6} = 816.$$

A knight can perform moves of the form $(\pm 1, \pm 2)$ and $(\pm 2, \pm 1)$, so

$$K = 8 \cdot (9-2)(9-1) = 448.$$

Therefore,

$$B + K + R = 816 + 448 + 1296 = \boxed{2560}.$$

TR 10: There is an infinite row of cars. Each car independently has a $\frac{1}{3}$ chance of being high quality and a $\frac{2}{3}$ chance of being low quality. John goes down the row of cars buying

high-quality cars. However, after John sees 3 low-quality cars, he gives up on buying additional cars. The probability that he buys at least 5 cars before giving up can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{254}$

Solution: We find a general formula for the probability that John buys k cars before giving up. John must see the third low-quality car after buying k high-quality cars, and the number of ways to order this is $\binom{k+2}{2}$. The probability of getting k high-quality cars and 3 low-quality cars is

$$\left(\frac{1}{3}\right)^k \cdot \left(\frac{2}{3}\right)^3,$$

so the probability that John buys k cars before giving up is

$$\frac{\binom{k+2}{2} \cdot 2^3}{3^{k+3}}.$$

The complement of the desired event is that John gives up after buying fewer than 5 cars, so substituting $k = 0, 1, 2, 3, 4$ and summing gives $\frac{232}{243}$. Therefore, the desired probability is

$$\frac{11}{243} \implies 11 + 243 = \boxed{254}.$$

TR 11: Let $ABCD$ be a cyclic quadrilateral such that AC is a diameter. Let P be the orthocenter of ABD. Let X be the intersection of lines AB and CD, and let Y be the intersection of lines AD and BC. If $AB = 8$, $BC = 1$, and $CD = 4$, then $\frac{[CBPD]}{[AXY]}$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{191}$

Solution:

Since AC is a diameter, $\angle ABC$ and $\angle ADC$ are right angles. Thus, since $\angle XBY = \angle XDY = 90°$, quadrilateral $BXYD$ is cyclic. It is well known that $BCDP$ is a parallelogram, which can be verified by angle chasing. From the sine area formula, we have

$$[CBPD] = 2 \cdot \frac{1}{2} \cdot CB \cdot CD \cdot \sin\angle BCD \quad \text{and} \quad [AXY] = \frac{1}{2} \cdot AX \cdot AY \cdot \sin\angle BAD.$$

Since $\sin\angle BCD = \sin\angle BAD$, the ratio of the areas is

$$2 \cdot \frac{CB \cdot CD}{AX \cdot AY}.$$

Since the diameter has length $\sqrt{65}$, we have $AD = 7$. Let $XC = a$. By power of a point at X, we get

$$\begin{aligned} a(a+4) &= \sqrt{a^2-1}\left(\sqrt{a^2-1}+8\right), \\ a^2+4a &= a^2-1+8\sqrt{a^2-1}, \\ 16a^2+8a+1 &= 64a^2-64, \end{aligned}$$

which gives $XC = \frac{5}{4}$. Since $CX \cdot CD = CY \cdot CB$, it follows that $CY = 5$. Thus $BX = \frac{3}{4}$ and $DY = 3$. Therefore, the ratio is

$$2 \cdot \frac{1 \cdot 4}{\frac{35}{4} \cdot 10} = \frac{16}{175} \implies 16 + 175 = \boxed{191}.$$

TR 12: Let T be the set of rationals of the form $\frac{a}{2^b}$ for nonnegative a and b. Define the function $f\colon T \to \mathbb{Z}$ such that, for $t = \frac{a}{2^b}$ such that b is minimal, we have that

$$f\left(\frac{a}{2^b}\right) = \begin{cases} 1 & b = 0 \\ f\left(\frac{a-1}{2^b}\right) + f\left(\frac{a+1}{2^b}\right) & b \neq 0 \end{cases}$$

The value of

$$\sum_{i=0}^{2^{10}-1} \frac{f\left(\frac{i+1}{2^{10}}\right)}{f\left(\frac{i}{2^{10}}\right)}$$

can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{3073}$

Solution: Define

$$S_n = \sum_{i=0}^{2^n-1} \frac{f\left(\frac{i+1}{2^n}\right)}{f\left(\frac{i}{2^n}\right)},$$

for $n \geq 1$. We claim that

$$S_n = \frac{5}{2} + 3 \cdot (2^{n-1} - 1)$$

by induction. The base case $S_1 = \frac{5}{2}$ holds. By symmetry, we have $f(t) = f(1-t)$ for $0 \leq t \leq 1$. Thus, for $n \geq 1$, we may pair terms to write

$$S_n = \sum_{i=0}^{2^{n-1}-1} \frac{f\left(\frac{i+1}{2^n}\right)}{f\left(\frac{i}{2^n}\right)} + \frac{f\left(\frac{i}{2^n}\right)}{f\left(\frac{i+1}{2^n}\right)}.$$

Then

$$\begin{aligned} S_{n+1} &= \sum_{i=0}^{2^n-1} \frac{f\left(\frac{i+1}{2^{n+1}}\right)}{f\left(\frac{i}{2^{n+1}}\right)} + \frac{f\left(\frac{i}{2^{n+1}}\right)}{f\left(\frac{i+1}{2^{n+1}}\right)} \\ &= \sum_{i=0}^{2^{n-1}-1} \frac{f\left(\frac{2i+1}{2^{n+1}}\right)}{f\left(\frac{2i}{2^{n+1}}\right)} + \frac{f\left(\frac{2i}{2^{n+1}}\right)}{f\left(\frac{2i+1}{2^{n+1}}\right)} + \frac{f\left(\frac{2i+1}{2^{n+1}}\right)}{f\left(\frac{2i+2}{2^{n+1}}\right)} + \frac{f\left(\frac{2i+2}{2^{n+1}}\right)}{f\left(\frac{2i+1}{2^{n+1}}\right)}. \end{aligned}$$

Since $f\left(\frac{2i+1}{2^{n+1}}\right) = f\left(\frac{2i}{2^{n+1}}\right) + f\left(\frac{2i+2}{2^{n+1}}\right)$, this becomes

$$\begin{aligned} S_{n+1} &= \sum_{i=0}^{2^{n-1}-1} \left(3 + \frac{f\left(\frac{2i+2}{2^{n+1}}\right)}{f\left(\frac{2i}{2^{n+1}}\right)} + \frac{f\left(\frac{2i}{2^{n+1}}\right)}{f\left(\frac{2i+2}{2^{n+1}}\right)}\right) \\ &= 3 \cdot 2^{n-1} + S_n. \end{aligned}$$

This completes the induction. Thus,

$$S_{10} = \frac{5}{2} + 3 \cdot 511 = \frac{3071}{2} \implies 3071 + 2 = \boxed{3073}.$$

TR 13: Let $D(n)$ denote the product of all divisors of n. Let $P(i,j)$ denote the set of all positive integers that are both a multiple of i and a factor of j. Let

$$F(x) = -\sqrt{\left|\log_{10}\left(\frac{D(10^x)}{\prod_{\omega\in P(10,10^{x+1})}\omega}\right)\right|} \quad \text{and} \quad G(n) = \sqrt[n-1]{\prod_{i=2}^{n} 10^{F(i)}}.$$

The value of $\sum_{k=2}^{\infty} G(k)$ can be expressed as $\frac{a+b\sqrt{c}}{d}$, where a, b, c, and d are positive integers, c is not divisible by the square of any prime, and $\gcd(a,b,d) = 1$. Find the value of $a+b+c+d$.

Answer: $\boxed{9021}$

Solution: Let $\tau(n)$ denote the number of positive divisors of n. Then $D(n) = n^{\tau(n)/2}$, which follows by pairing divisors whose product is n. Thus,

$$D(10^x) = (10^x)^{\frac{(x+1)^2}{2}} = 10^{\frac{x(x+1)^2}{2}}.$$

The set $P(10, 10^{x+1})$ consists of the $(x+1)^2$ integers of the form $10m$ where $m \mid 10^x$, so the product of the elements of P is

$$\prod_{m|10^x} 10m = 10^{(x+1)^2} \cdot D(10^x).$$

Hence,

$$\begin{aligned} F(x) &= -\sqrt{\left|\log_{10}\left(\frac{D(10^x)}{10^{(x+1)^2}\cdot D(10^x)}\right)\right|} \\ &= -\sqrt{\left|\log_{10}\left(10^{-(x+1)^2}\right)\right|} \\ &= -x-1. \end{aligned}$$

Now,

$$G(n) = \sqrt[n-1]{\prod_{i=2}^{n} 10^{-i-1}} = \sqrt[n-1]{\left(10^{\frac{-(n+4)}{2}}\right)^{n-1}} = 10^{\frac{-(n+4)}{2}}.$$

Finally,

$$\sum_{k=2}^{\infty} 10^{\frac{-(k+4)}{2}} = \frac{1}{1000}\left(\frac{1}{1-\frac{1}{\sqrt{10}}}\right) = \frac{10+\sqrt{10}}{9000} \implies 10+1+10+9000 = \boxed{9021}.$$

TR 14: For how many primes $p < 100$ does there exist a positive integer k such that p divides $10^k + 1$?

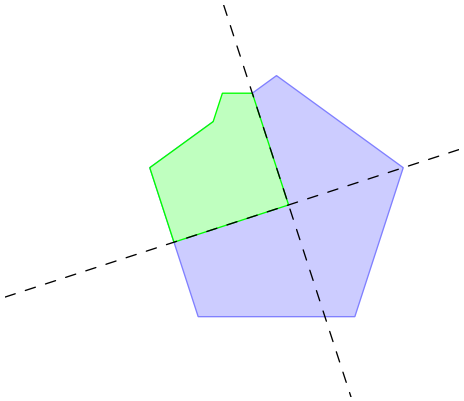
Answer: $\boxed{13}$

Solution: The primes 2 and 5 do not work, since $10^k + 1 \equiv 1 \pmod{p}$ for both. For any other prime $p < 100$, let d be the order of 10 modulo p. Since -1 is the unique element of order 2 modulo p, some power of 10 equals -1 if and only if d is even. Write $p-1 = 2^s q$, where q is odd. Since $d \mid p-1$, the order d is odd if and only if $10^q \equiv 1 \pmod{p}$. We have the following table of q and $10^q \bmod p$ for each prime:

p	q	$10^q \bmod p$	p	q	$10^q \bmod p$
3	1	1	41	5	1
7	3	6	43	21	1
11	5	10	47	23	46
13	3	12	53	13	1
17	1	10	59	29	58
19	9	18	61	15	50
23	11	22	67	33	1
29	7	17	71	35	1
31	15	1	73	9	10
37	9	1	79	39	1
			83	41	1
			89	11	55
			97	3	30

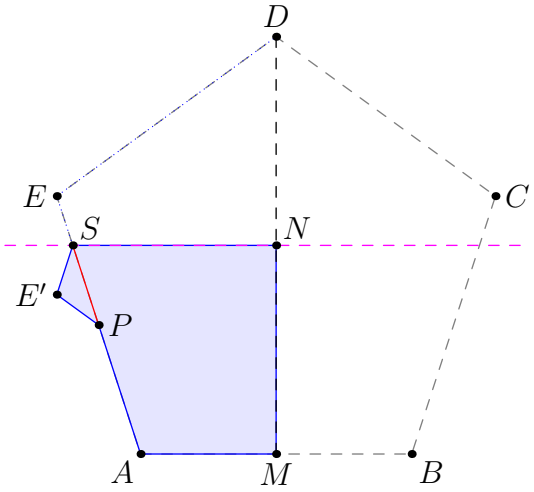
Exactly 10 of the 23 primes in the table have odd order, so the answer is $23 - 10 = \boxed{13}$.

TR 15: Consider a piece of paper in the shape of a regular pentagon with side length 2. We fold it in half. We then fold it again so that the two endpoints of its longest side coincide. The area of the folded figure can be expressed as $\frac{1}{a}\sqrt{b + c\sqrt{5}}$, where a, b, c are positive integers with a minimized. Find $a + b + c$.



Answer: $\boxed{235}$

Solution:



Let the pentagon be $ABCDE$, and let M be the midpoint of AB. The first fold is along DM, leaving quadrilateral $AMDE$ with $AM = 1$ and $AE = ED = 2$. It is easy to compute that $DM = \sqrt{5+2\sqrt{5}}$, so DM is the longest side. The second fold is along the perpendicular bisector of DM, which meets DM at N and AE at S. Since $\angle EAB = 108^\circ$, the point S lies a horizontal distance of

$$\frac{DM}{2}\tan 18^\circ = \frac{\sqrt{5+2\sqrt{5}}}{2}\cdot\sqrt{\frac{5-2\sqrt{5}}{5}} = \frac{1}{2}$$

to the left of A. So, $SN = \frac{3}{2}$, and trapezoid $AMNS$ has area

$$\frac{1}{2}\left(1+\frac{3}{2}\right)\cdot\frac{DM}{2} = \frac{5}{8}\sqrt{5+2\sqrt{5}}.$$

Now, the fold sends D to M, E to a point E', and $DNSE$ to $MNSE'$. Let ME' meet AE at P. The only part of $MNSE'$ outside $AMNS$ is triangle SPE'. From the Pythagorean Theorem,

$$AS = \sqrt{\frac{1}{4}+\frac{5+2\sqrt{5}}{4}} = \frac{1+\sqrt{5}}{2},$$

so $SE' = SE = 2 - AS = \frac{3-\sqrt{5}}{2}$. Since $\angle E'SN = \angle ESN = 108^\circ$ and $\angle ASN = 72^\circ$, we have $\angle E'SP = 36^\circ$. Also, $\angle SE'P = \angle SED = 108^\circ$, so $\angle SPE' = 36^\circ$ and $E'P = E'S$. So, from the sine area formula,

$$[SPE'] = \frac{1}{2}\left(\frac{3-\sqrt{5}}{2}\right)^2\sin 108^\circ = \frac{7-3\sqrt{5}}{2}\cdot\frac{\sqrt{10+2\sqrt{5}}}{8}.$$

Since $\frac{10+2\sqrt{5}}{5+2\sqrt{5}} = 6-2\sqrt{5} = \left(\sqrt{5}-1\right)^2$, we have $\sqrt{10+2\sqrt{5}} = \left(\sqrt{5}-1\right)\sqrt{5+2\sqrt{5}}$, giving

$$[SPE'] = \frac{(7-3\sqrt{5})(\sqrt{5}-1)}{16}\sqrt{5+2\sqrt{5}} = \frac{5\sqrt{5}-11}{8}\sqrt{5+2\sqrt{5}}.$$

Thus, the area of the folded figure is

$$\begin{aligned}\frac{5}{8}\sqrt{5+2\sqrt{5}}+\frac{5\sqrt{5}-11}{8}\sqrt{5+2\sqrt{5}} &= \frac{5\sqrt{5}-6}{8}\sqrt{5+2\sqrt{5}}\\ &= \frac{1}{8}\sqrt{(161-60\sqrt{5})(5+2\sqrt{5})}\\ &= \frac{1}{8}\sqrt{205+22\sqrt{5}}.\end{aligned}$$

Since b and c are integers, a^2 must be a multiple of 64, so $a = 8$ is minimal, and the answer is $8+205+22 = \boxed{235}$.